

Enrico Tazzer

✉ enri.tazzer@gmail.com 📞 +39 345 5340795 📍 Rivamonte Agordino (BL), Italy
🐙 github.com/enricotazzer 🔗 linkedin.com/in/enrico-tazzer 🌐 enricotazzer.github.io

SUMMARY

MSc student in the double-degree programme in Artificial Intelligence and Cybersecurity at the Università degli Studi di Udine and the Alpen-Adria-Universität Klagenfurt, with a current master's weighted average of 28.70/30 and an expected final grade of 110/110. I build machine learning systems for medicine end to end, deep learning and neuro-symbolic models for medical imaging and clinical NLP, the evaluation that says whether they actually work, and the interface that makes them usable. My work is organised around keeping claims inside what the evidence supports: explicit baselines, calibration, ablations, subgroup fairness audits, and negative results reported rather than buried. Naturally curious and hands-on, eager to take on new problems within and beyond the core field. Available from October 2026.

EDUCATION

M.Sc. in Artificial Intelligence and Cybersecurity	Oct. 2024 – Oct. 2026 (expected)
<i>Double degree programme — weighted average 28.70/30</i>	<i>Università degli Studi di Udine (IT) & Universität Klagenfurt (AT)</i>
<i>Erasmus semester at Universität Klagenfurt, Klagenfurt am Wörthersee (AT)</i>	<i>Oct. 2025 – Feb. 2026</i>
B.Sc. in Computer Science	Sep. 2021 – Oct. 2024
<i>Final grade: 97/110</i>	<i>Università degli Studi di Udine, Udine (IT)</i>

RELEVANT COURSEWORK

Artificial Intelligence & Machine Learning: Deep Learning, Computer Vision, Recommender Systems, Automated Reasoning, Verification & Validation Techniques for AI & Cybersecurity
Computer Science Fundamentals: Data Structures & Algorithms, Object-Oriented Programming, Discrete Mathematics, Linear Algebra, Probability & Statistics, Complexity and Information Theory

TECHNICAL SKILLS

Languages: Python, C/C++, Java, C#, TypeScript/JavaScript, SQL, HTML/CSS, L^AT_EX
ML & Deep Learning: PyTorch, PyTorch Geometric, timm, captum, TensorFlow, Keras, Scikit-learn, lifelines, HuggingFace, vLLM, LangChain, LangGraph
Backend & Data: FastAPI, PostgreSQL/pgvector, SQLAlchemy, Alembic, Docker, React, Streamlit, NumPy, Pandas, Matplotlib
Symbolic AI & Planning: Clingo/eclingo, MiniZinc, PDDL, Fast Downward, plasp
Tooling: Git, pytest, ruff, Unity, Kaggle GPU

RESEARCH EXPERIENCE

AI-Assisted Extraction of Opportunistic Findings From Oncologic CT Reports Using Multimodal Large Language Models
HuggingFace, PyTorch — Master's thesis (in progress) Mar. 2026 – Present

- Benchmarking state-of-the-art open-source medical MLLMs (MedGemma, LLaVA-Med) for extracting opportunistic findings from oncologic CT studies.
- Implemented and compared several image-input strategies, plus an image-only inference pipeline that generates the textual report directly from the scan.
- Extensive quantitative evaluation of each approach adopted.

Multitask Learning and Neuro-Symbolic Reasoning: A Comparative Exploration for Skin Lesion Segmentation and Classification
Python, PyTorch, Clingo | [GitHub](#) Nov. 2025 – Jun. 2026

- Multitask learning for joint skin lesion classification and segmentation, compared against single-task baselines with quantitative and statistical significance evaluation.

- Neuro-symbolic reasoning layer over the network outputs to improve explainability and effectiveness, cross-evaluated against the purely neural approach.

Italian Medical Report Encoding by NLP — Bachelor's thesis

Python, PyTorch, HuggingFace, Scikit-Learn, Tensorflow | [GitHub](#)

Jun. 2024 – Sep. 2024

- NLP pipeline for extracting structured information from free-text Italian clinical reports and mapping it to SNOMED-3 codes.
- Comparison between fine-tuned BERT and classical ML baselines for the classification task.

SELECTED PROJECTS

Chest X-Ray Triage Assistant with Explainability

PyTorch, timm, captum, Streamlit | [GitHub](#)

Jul. 2026 – Aug. 2026

- Multi-label DenseNet-121 on NIH ChestX-ray14 (112,120 images, official split): **0.802 mean test AUC** with bootstrap 95% confidence intervals, above the Wang et al. (2017) benchmark (≈ 0.75); ResNet-50 trained as a comparison run.
- Grad-CAM explanations validated against random controls and occlusion tests (pointing game 0.374 vs 0.101 control), exposing that the flagship emergent finding rests on a lateralised whole-lung cue rather than the apical pleural line.
- Reading-queue simulation: model-ordered triage cuts urgent-case mean wait by **54%** ($15.9 \rightarrow 7.4$ min) and p95 by 59% against arrival order, with the cost to routine cases quoted next to the win.
- Subgroup fairness audit over sex, age and view position, surfacing a significant AP-vs-PA AUC gap ($+0.026$); 105 tests, Streamlit demo, training on Kaggle's free GPU.

Oracle: Oncology Reconstruction And Clinical Learning Engine

Python, PyTorch, Three.js | [GitHub](#) | [Project page](#)

Jan. 2026 – Jun. 2026

- End-to-end deep learning framework for brain tumor analysis from multimodal MRI, integrating segmentation, sparse-to-dense 3D reconstruction and physics-informed growth prediction.
- nnU-Net 2D segmentation with TTA post-processing; 2.5D generator fine-tuned against a 3D GAN discriminator for volume reconstruction, with an interactive Three.js brain viewer (marching cubes with Taubin smoothing, before/after volume toggle, slice clipping).
- PINN to predict brain tumor evolution over time.

Lung Tumor Segmentation Benchmark | *PyTorch, U-Mamba* | [GitHub](#)

Sep. 2025 – Mar. 2026

- Systematic benchmark of state-of-the-art ML and DL segmentation models plus the cutting-edge U-Mamba, all evaluated under identical settings for a fair comparison.
- Analysis of the main architectures and their effectiveness in the medical domain.

WasteWise — household food-waste and grocery-spend system

FastAPI, PostgreSQL/pgvector, React, Docker, lifelines | [GitHub](#)

Aug. 2026

- Full-stack system built end to end: FastAPI + PostgreSQL/pgvector + React 19/TypeScript, Dockerised, **378 tests**, with every reported figure regenerated from the database rather than written down.
- Four-tier barcode resolution pipeline (local cache, Open Food Facts, LLM categorisation via Groq with Gemini failover, manual bind), each tier degrading into the next so a scan never dead-ends.
- Survival analysis (Cox PH, gradient boosting) for P(consumed before expiry) behind an explicit data gate, with model selection enforced in code: the v1 model lost to the v0 heuristic, so the scorer refuses to load it.
- Recipe recommender over a self-crawled, robots.txt-respecting corpus of 1,666 Italian recipes embedded on CPU for pgvector reranking.

TacticalGraph — graph learning on Serie A event data

PyTorch Geometric, GraphSAGE, Streamlit | [GitHub](#)

Aug. 2026

- Harmonised **1.26M actions across two providers** (StatsBomb and Wyscout, 760 matches) through the SPADL representation, with a leakage guard enforced mechanically at every feature-matrix boundary.
- GraphSAGE embeddings of passing networks for player functional role: **ARI 0.516 vs 0.051** for the classical-centrality baseline, with a three-variant feature ablation over three seeds.
- Quantified the season/provider confound (per-match action-rate ratios up to $296\times$) and reported the negative result honestly — passing topology adds only $+1.1$ to $+1.5$ pp over pitch position.

Coursework Projects | *Clingo/eclingo, PDDL, Fast Downward, plasp, Unity, C#*

2026

- **Hospital Room Assignment Planning** — PDDL domain modelling patient-to-room assignment under medical equipment constraints; universal reversibility verified by translating instances to ASP via plasp and checking them with eclingo and clingo. [GitHub](#)
- **Chrono Fracture** — puzzle game built in Unity around time-shift mechanics and an immersive story (videogame programming exam project).

LANGUAGES

Italian: native speaker | **English:** fluent (professional working proficiency; no formal certification)

EXTRA ACTIVITIES

Internship at Luxottica S.p.A. | *IT HelpDesk office, Agordo (BL) establishment*

Jul. 2019